

Let's observe how both methods work. Using this pilot data, we can easily visualize the impact of RMANOVA as it deletes data that is not fully representative of all events (data points). This pilot study has 10 individuals with a total event of four clinical measures.

| Obs | Patient_ID | Group     | Age | Week_0 | Week_4 | Week_8 | Week_12 |
|-----|------------|-----------|-----|--------|--------|--------|---------|
| 1   | P001       | Treatment | 45  | 82.5   | 78.1   | 70.4   | 62.1    |
| 2   | P002       | Treatment | 52  | 79.0   | 74.3   | 68.0   | 60.5    |
| 3   | P003       | Control   | 38  | 85.2   | 84.0   | 86.1   | 84.9    |
| 4   | P004       | Treatment | 61  | 91.4   | 85.0   | .      | 71.3    |
| 5   | P005       | Control   | 49  | 75.0   | 76.2   | 74.8   | 77.0    |
| 6   | P006       | Treatment | 34  | 88.1   | 80.2   | 73.5   | 65.4    |
| 7   | P007       | Control   | 55  | 80.3   | 81.5   | 79.9   | .       |
| 8   | P008       | Control   | 42  | 86.0   | 85.5   | 87.2   | 86.5    |
| 9   | P009       | Treatment | 47  | 93.7   | 88.4   | 81.0   | 72.8    |
| 10  | P010       | Control   | 50  | 78.5   | .      | 79.0   | 80.2    |

For RMANOVA we can see the patients 4, 7 and 10 have missing data for event week 8, week 12 and week 4. Based on the methodology of RMANOVA these 3 patients will be eliminated from the sample leading to a sample size of 7.

The syntax for RMANOVA

```
proc glm data=clinical_trial;
  class Group;
  model Week_0 Week_4 Week_8 Week_12 = Group / nouni;
  repeated Time 4 / nom;
  title "Method A: Repeated Measures ANOVA (Wide Data)";
run;
quit;
```

| Method A: Repeated Measures ANOVA (Wide Data)                                                                   |    |             |             |         |        |  |
|-----------------------------------------------------------------------------------------------------------------|----|-------------|-------------|---------|--------|--|
| The GLM Procedure<br>Repeated Measures Analysis of Variance<br>Tests of Hypotheses for Between Subjects Effects |    |             |             |         |        |  |
| Source                                                                                                          | DF | Type III SS | Mean Square | F Value | Pr > F |  |
| Group                                                                                                           | 1  | 267.1433333 | 267.1433333 | 1.98    | 0.2186 |  |
| Error                                                                                                           | 5  | 675.4316667 | 135.0863333 |         |        |  |

  

| Method A: Repeated Measures ANOVA (Wide Data)                                                                            |    |             |             |         |        |                 |
|--------------------------------------------------------------------------------------------------------------------------|----|-------------|-------------|---------|--------|-----------------|
| The GLM Procedure<br>Repeated Measures Analysis of Variance<br>Univariate Tests of Hypotheses for Within Subject Effects |    |             |             |         |        |                 |
| Source                                                                                                                   | DF | Type III SS | Mean Square | F Value | Pr > F | Adj Pr > F      |
|                                                                                                                          |    |             |             |         |        | G - G   H-F-L   |
| Time                                                                                                                     | 3  | 374.4107143 | 124.8035714 | 153.45  | <.0001 | <.0001   <.0001 |
| Time*Group                                                                                                               | 3  | 446.6584286 | 148.8854762 | 183.06  | <.0001 | <.0001   <.0001 |
| Error(Time)                                                                                                              | 15 | 12.2000000  | 0.8133333   |         |        |                 |

  

|                              |        |
|------------------------------|--------|
| Greenhouse-Geisser Epsilon   | 0.7327 |
| Huynh-Feldt-Lecoutre Epsilon | 1.3309 |

The F value is the overall outlook that shows that comparing the treatment and control group effect is not significant at the (F value =1.98,  $p=0.2186$ ). The time effect is significant at (F-value =153.45,  $p<0.0001$ ). The interaction effect of time and group is significant at (F-value of 183.06,  $p<0.0001$ ).

The Epsilon is the vital correction that adjusts the degrees of freedom ( $df$ ) when data violate the assumption of sphericity. According to Ellen R. Girden (1992) (0.75-1.00), an epsilon value below 0.75 indicates a severe violation of sphericity. Because the calculated Greenhouse-Geisser epsilon is 0.7327 the assumption of sphericity is violated, and the Greenhouse-Geisser adjusted p-value ( $p < 0.0001$ ) should be reported to protect against Type I errors (false positives).